

GUMPOLDSKIRCHEN, Austria

WMO: 110820

Lat: 48.0358N

Lon: 16.2808E

Elev: 212

StdP: 98.80

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.4	-7.2	-13.9	1.2	-8.1	-11.3	1.5	-5.6	10.8	3.2	9.1	4.3	1.5	340	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	31.9	21.5	30.0	20.7	28.2	19.8	22.2	30.0	21.3	28.6	20.4	27.0	2.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.7	26.0	18.6	13.8	24.9	17.8	13.1	24.3	66.0	30.1	62.5	28.8	59.5	27.0	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.6	7.2	6.1	-11.9	34.5	3.0	2.4	-14.1	36.3	-15.8	37.7	-17.5	39.0	-19.6	40.7		
			-12.6	23.6	2.8	0.9	-14.6	24.3	-16.3	24.8	-17.9	25.3	-20.0	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	0.5	2.5	6.4	11.6	15.9	19.6	21.5	21.1	16.1	11.0	6.1	1.6
	DBStd	8.35	4.56	4.74	4.31	4.08	3.54	3.53	3.25	3.18	3.35	3.85	3.99	4.30
	HDD10.0	1089	295	212	124	31	2	0	0	0	1	35	126	263
	HDD18.3	2901	552	443	369	203	92	28	9	9	82	228	366	520
	CDD10.0	1530	1	3	14	79	183	287	356	344	185	66	11	1
	CDD18.3	301	0	0	0	2	16	64	107	96	16	1	0	0
	CDH23.3	2422	0	0	0	19	121	512	887	778	99	4	0	0
	CDH26.7	752	0	0	0	2	16	143	303	271	17	0	0	0
(14) Wind	WSAvg	2.5	2.5	2.5	2.6	2.7	2.8	2.6	2.8	2.4	2.4	2.0	2.1	2.3
(15) Precipitation	PrecAvg	649	35	32	48	43	71	78	78	70	66	45	44	38
	PrecMax	865	87	74	119	99	168	179	226	169	201	97	86	69
	PrecMin	433	3	0	14	1	18	23	8	15	20	4	0	11
	PrecStd	112	20	22	28	27	36	38	43	40	41	24	19	17
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	16.5	20.2	25.7	28.5	32.5	34.4	34.9	28.9	23.4	18.6	13.9
		MCWB	9.0	9.3	12.0	16.0	19.4	21.5	22.2	21.2	19.2	16.4	13.0	9.5
	2%	DB	11.7	13.6	17.5	22.8	26.3	30.2	31.9	32.0	26.1	20.7	16.1	11.6
		MCWB	7.8	8.5	10.7	14.7	18.1	21.0	21.5	21.3	18.3	15.2	11.8	7.7
	5%	DB	9.3	11.4	15.1	20.6	24.5	28.2	29.9	29.7	23.8	18.8	13.8	9.2
		MCWB	6.1	7.3	9.5	13.5	17.2	19.9	20.6	20.6	17.2	14.2	10.4	6.2
	10%	DB	7.2	9.1	13.0	18.6	22.6	26.3	28.1	27.5	21.9	16.9	11.8	7.3
		MCWB	4.6	5.9	8.5	12.3	16.0	18.9	19.8	19.5	16.3	13.2	9.1	4.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	10.2	12.6	16.6	20.2	23.2	23.7	23.1	20.1	17.0	13.7	9.7
		MCDB	13.5	15.1	18.7	24.4	27.0	30.2	31.7	31.7	27.3	21.7	17.3	13.0
	2%	WB	8.0	8.7	11.1	15.1	18.9	21.6	22.2	21.9	18.9	15.8	12.2	8.0
		MCDB	11.3	13.1	16.4	22.2	25.4	28.7	30.1	30.0	24.9	19.7	15.6	11.2
	5%	WB	6.3	7.5	10.0	13.9	17.7	20.5	21.3	21.0	17.8	14.7	10.7	6.4
		MCDB	9.0	11.2	14.5	19.7	23.4	27.0	28.6	28.5	22.7	18.0	13.5	8.9
	10%	WB	4.7	6.1	8.9	12.8	16.6	19.5	20.4	20.0	16.9	13.6	9.4	5.0
		MCDB	7.0	9.1	12.6	17.8	21.8	25.4	26.8	26.5	21.0	16.4	11.5	7.1
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	6.2	7.6	9.1	9.0	9.1	9.3	9.7	8.1	7.5	5.3	4.8
		MCDBR	8.1	9.4	11.8	12.9	12.7	12.6	12.9	13.6	11.8	10.4	8.3	8.3
	5% WB	MCWBR	5.3	5.9	6.6	6.5	6.1	5.4	4.8	5.1	5.3	5.7	5.2	5.6
		MCWBR	7.7	8.8	11.0	12.0	11.9	11.9	11.9	12.7	11.0	9.6	7.8	7.7
(40) Clear-Sky Solar Irradiance	taub		0.317	0.342	0.382	0.427	0.426	0.431	0.437	0.437	0.408	0.377	0.347	0.316
		taud	2.385	2.341	2.261	2.167	2.217	2.241	2.226	2.241	2.290	2.349	2.386	2.413
	Ebn at Noon		750	808	827	822	839	835	823	805	790	754	701	703
		Edh at Noon	69	90	114	138	136	134	134	127	110	89	70	60
(44) All-Sky Solar Radiation	RadAvg		1.02	1.84	3.00	4.44	5.23	5.70	5.52	4.92	3.52	2.19	1.14	0.80
	RadStd		0.12	0.24	0.38	0.45	0.56	0.43	0.50	0.48	0.43	0.24	0.12	0.09

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.79	N/A	N/A	+1.19	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (91 neighbors)	+0.56	N/A	N/A	+0.84	+0.45	+0.38	-120	-160	+85	+43

Nomenclature: See separate page